LAB # 6b
Implementation of break and continue keyword in java

Objective

To gain familiarity with the Implementation of break and continue keyword in java.

Lab Goals

To provide working insight into the usage of Implementation of break and continue keyword in java:

a. Implementation of break keyword in java.
b. Implementation of continue keyword in java

Introduction to Break and Continue in Java

In Java, "break" and "continue" are control flow statements used within loops to alter the flow of execution.

break Statement:

When a break statement is encountered inside a loop, the loop immediately terminates, and the program continues with the next statement after the loop.

Example#1

```
public class BreakExample {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            if (i == 3) {
                break; // terminate the loop when i equals 3
            }
            System.out.println(i);
        }
        System.out.println("Loop ended.");
    }
}
```

Output

```
1
2
Loop ended.
```

Example#2

```
public class Main {
    public static void main(String[] args) {
        for (int i = 0; i < 10; i++) {
            if (i == 5) {
                break; // Exit the loop when i equals 5
            }
            System.out.println(i);
        }
    }
}
```

Output

```
0
1
2
3
4
```

continue Keyword

The continue statement is used to skip the remaining code inside a loop for the current iteration and proceed with the next iteration of the loop. It is typically used when you want to skip some specific condition but continue with the rest of the loop iterations.

Example#1

```
public class ContinueExample {
    public static void main(String[] args) {
        for (int i = 1; i <= 5; i++) {
            if (i == 3) {
                continue; // skip printing when i equals 3
            }
            System.out.println(i);
        }
        System.out.println("Loop ended.");
    }
}
```

Output

```
1
2
4
5
Loop ended.
```

Example#2

```
public class Main {
    public static void main(String[] args) {
        for (int i = 0; i < 10; i++) {
            if (i % 2 == 0) {
                continue; // Skip even numbers
            }
            System.out.println(i);
        }
    }
}
```

Output

```
1
3
5
7
9
```

Exercise

A. Create a Java program that uses nested loops, the break and continue keywords, and a given integer n to identify the first instance of a prime number bigger than that number.

B. Write a Java program to search for a specific character in a given string. If found, print "**Character found at index**" followed by its index. If not found, print "**Character not found**".

C. Write a Java program to print the first 10 numbers of the Fibonacci sequence, skipping numbers that are divisible by 3.

D. Write a Java program to find the first non-repeated character in a given string. If all characters are repeated, print "**All characters are repeated**".